

Dr. Émile Emery

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Research Profile

Statistical physicist of complex networks. I **reconstruct complex networks** from large, heterogeneous and incomplete **data sets**, explain the resulting structures with parsimonious generative rules, and **model the dynamical processes** that run on them. The systems studied so far are infrastructural, ecological and social.

Education

- 2026** **French National Qualification for Associate Professor (CNU)**, sections 28 (Condensed Matter and Materials) and 29 (Elementary Constituents)
- 2022–2025** **PhD in Theoretical and Numerical Physics**, defended on 8 October 2025
Université Paris-Saclay / CEA IRAMIS–SPEC, Gif-sur-Yvette
Statistical physics and resource investments in networks for the energy transition
Supervisors: S. Aumaître and H. Bercegol. NNT: 2025UPASP078.
- 2021–2022** **Master’s Year 2 — ICFP** (*Theoretical Physics* programme),
École Normale Supérieure, Paris
- 2020–2021** **Master’s Year 1 — High Energy Physics**,
École Polytechnique, Palaiseau
- 2017–2020** **Magistère in Physics** (selective undergraduate programme), Université Paris Cité.
2nd year interrupted in 2018–2019, see Other Training.
- 2015–2017** **Preparatory classes MPSI-MP**, top-ranked in physics (MP), Lycée Chaptal, Paris

Research Positions

- Oct. 2025–Sept. 2026** **Postdoctoral Researcher**, Université Paris Cité
Paris Interdisciplinary Energy Research Institute (LIED/PIERI).
Supervisor: José Halloy. Funded by PEPR Recyclage.
Reconstruction and modelling of distribution networks: geospatial data assembly, scaling inference, topology estimation.
- Oct. 2022–Sept. 2025** **PhD Researcher**, CEA IRAMIS–SPEC, Gif-sur-Yvette
SPHYNX Group. Supervisors: Sébastien Aumaître, Hervé Bercegol.
Funded by CEA FOCUS TIME.
Statistical physics of large empirical networks: network reconstruction, generative growth models, optimisation of processes on network structures.
- Apr.–Aug. 2022** **Research Intern**, Raman Research Institute (RRI), Bangalore, India
Quantum Gravity Group. Supervisor: Sumati Surya.
Analytical work on causal set theory (graph theory).
- Mar.–Jul. 2021** **Research Intern**, Institut d’Astrophysique de Paris (IAP)
Cosmology and Gravitation Group. Supervisor: Guillaume Faye.
Perturbative computations (Mathematica) on compact binary systems.
- May–Jul. 2018** **Research Intern**, Astroparticles & Cosmology (APC), Paris
Cosmology Group. Supervisor: Cyrille Rosset.
Python-based image analysis pipeline for the Euclid project.

Publications

PhD thesis

Emery É. *Statistical physics and resource investments in networks for the energy transition.* PhD thesis, Université Paris-Saclay, defended on 8 October 2025. NNT: 2025UPASP078.

Peer-reviewed journal articles

Emery É., Le Bihan J., Halloy J. *Connecting electrical grid length and material stock to population density: comparison of a French-calibrated scaling with 35 electrified countries.* **Physical Review X Energy**, 2026. doi:10.1103/2xgs-8821

→ *Inferring what the data do not record: population density predicts undocumented network properties.*

Emery É., Aumaître S., Bercegol H. *Power grid network growth model based on rural–urban differentiation.* **Physical Review X Energy** 4, no. 3, 033014, 2025. doi:10.1103/kysp-hgxt

→ *Sequential growth with rural–urban differentiation: nearest neighbours connection in rural areas, preferential attachment in cities. Reproduces the French transmission grid’s topology and dynamical Joule losses.*

Emery É., Bercegol H., Jonquères N., Aumaître S. *Complex network analysis of transmission networks preparing for the energy transition: application to the current French power grid.* **The European Physical Journal B** 97, no. 201, 2024. doi:10.1140/epjb/s10051-024-00837-7

→ *Full national transmission grid modeled from operator data: efficiency stems from the high-voltage mesh interconnecting thermal plants; integrating diffuse renewables will require topological adjustment.*

Preprints and manuscripts under revision

Caligiuri A., **Emery É.**, Ferreira L. *et al.* *Time-varying ecological interactions characterise equilibrium and stability.* Preprint, arXiv:2506.22123 [q-bio.PE], 2025.

→ *Environmental variation makes species interactions network change over time. We measured how this reshuffling widens or narrows the conditions under which species coexist.*

Azaiez A., **Emery É.** *Friend-of-friend generative model for group interactions.* Under revision at **Physical Review E**. hal-05487223, 2026.

→ *Generative model for higher-order social interaction data from a local attachment rule.*

Emery É., Aumaître S., Bercegol H. *Energy-optimal placement of storage systems in transmission grids.* Under revision at **Journal of Energy Storage**. arXiv:2603.03179, 2026.

→ *Optimising a dynamical process over a fixed topology: storage sited so that the energy it returns exceeds the energy invested in it.*

Manuscripts in preparation

Emery É., Presigny C., De Domenico M. *Universal generative modelling for inhomogeneous node distributions in spatial networks.*

→ *We replace the uniform node distribution that most spatial generative models assume with a self-similar one, and test it on urban, road, electrical and biological networks.*

Emery É., Halloy J. *Spatial scaling laws for hierarchical distribution networks.*

→ *Scaling exponents of typical distribution networks derived from first principles rather than fitted, and confirmed on the French electricity and gas grid.*

Emery É., Le Bihan J., Halloy J. *Resource investment in deployment and renewal of worldwide long-term energy grids.*

→ *The material metabolism of worldwide grids: the copper and aluminium locked in as stock, and the renewal flows that keep them standing, followed to 2100 under five population scenarios.*

Emery É. *A large-scale open-access dataset of competitive social networks (data paper).*

→ *Records of large-scale competitions turned into open network datasets: directed, weighted, temporal and higher-order.*

Skills

Statistical physics	Network analysis; generative models; scaling laws derived analytically; comparison against null models; universality across systems.
Machine learning	Neural networks built and trained in <code>tensorflow</code> ; MNIST digit classification implemented from scratch in <code>numpy</code> . Regression, clustering, model selection and cross-validation.
Network reconstruction	Large networks assembled and validated from incomplete and inconsistent sources; missing layers inferred rather than assumed.
Dynamical processes	Dynamics on networks modelled and simulated; evolution and stability derived analytically, including under time-varying couplings; processes optimised over a fixed topology.
Ecological interactions	Time-varying interaction networks inferred from ecological time series; rewiring quantified and its effect on coexistence assessed.
Metabolism	Energy and material resources balances written on graphs; stocks and renewal flows derived from network structure and projected under scenarios.
Programming	Python, L ^A T _E X, Mathematica, Bash, Git, CAML, SQL — Complex networks and hypergraphs: <code>networkx</code> , <code>xgi</code> . — Large-scale data processing and visualisation: <code>pandas</code> , <code>geopandas</code> . — Machine learning and regression: <code>tensorflow</code> , <code>scikit-learn</code> .
Software	Gephi, VS Code, GitHub, Zotero, Inkscape, Logseq, Activity Browser.
Reproducibility	Reproducible analysis workflows. Code and data released with the publications.
Open science	Training in Data Management Plans and open-science fundamentals.
Peer review	IOP Publishing Peer Review Excellence training.
Languages	French (native); English (fluent — publications, conferences and teaching); Italian and Spanish (working knowledge); Lojban (beginner).

Invited Talks and Conferences

Jun. 2026	<i>Journées françaises des Systèmes Complexes</i> , Lyon. <i>[Talk]</i>
Feb. 2026	Invited seminar, CoMuNe Lab, University of Padua , Italy. <i>[Invited talk]</i>
Jan. 2026	<i>Winter Workshop on Complex Systems</i> , Mallorca, Spain. <i>[Poster]</i>
Jan. 2026	Invited seminar, LIED, Université Paris Cité. <i>[Invited talk]</i>
Jul. 2025	<i>10th Mediterranean School of Complex Networks</i> , Syracuse, Italy. <i>[Talk]</i>
Jul. 2025	<i>Focus Time Seminar</i> , Grenoble. <i>[Talk + Poster]</i>
Jun. 2025	<i>NetSci 2025 — International School and Conference on Network Science</i> , Maastricht, Netherlands. <i>[Poster]</i>
Apr. 2025	<i>4th MOMENTOM International Congress</i> , Université Paris-Saclay. <i>[Talk]</i>
Jul. 2024	<i>Focus Time Seminar</i> , Aix-les-Bains. <i>[Talk + Poster]</i>
May 2024	Invited seminar, <i>Networks & Games</i> research group, Centre d'Économie de la Sorbonne, Paris. <i>[Invited talk]</i>
Mar. 2024	Doctoral school “Énergie & Recherche”, Roscoff. <i>[Talk]</i>
Jan. 2024	<i>NetSciX 2024 — International School and Conference on Network Science</i> , Venice, Italy. <i>[Talk]</i>
Oct. 2023	<i>Focus Time Seminar</i> , Aix-les-Bains. <i>[Talk]</i>
Oct. 2023	<i>Journée Systèmes et Matières Complexes</i> , LPS, Orsay. <i>[Talk]</i>
Jul. 2023	<i>Exergy Economics Workshop</i> , Chambéry. <i>[Talk]</i>
Mar. 2023	<i>3rd MOMENTOM International Congress</i> , Université Paris-Saclay. <i>[Poster]</i>
Nov. 2022	<i>Young Physicists' Meeting</i> , Collège de France, Paris. <i>[Talk]</i>

Awards and Distinctions

- 2026 Admissibility for an Associate Professor position, LIED/PIERI, Université Paris Cité
- 2026 Selected participant, *MSCA MaRaThoN* (intensive 3-day MSCA proposal training), University of Padua
- 2023 Best Poster Award, *3rd MOMENTOM International Congress*, Université Paris-Saclay

Scientific and Academic Responsibilities

- Reviewer for *Communications Earth & Environment* (Nature Portfolio), 2026.
- Co-organiser of the *Winter Workshop on Complex Systems 2027*, Briançon.
- Co-organiser of the *LIED Days 2026*, Université Paris Cité.
- Co-organiser of a symposium for the *LIED Manifesto* project 2026, Université Paris Cité.

Supervision

Spring 2026 — co-supervision of a Master's internship at LIED (with P. Chatzimpiros, H. Bercegol and S. Aumaître): *Modelling the railway development during the industrial revolution*.

Teaching

Period	Volume	Course / Institution / Level
Mar.–Jun. 2025	13.5 h TD equiv.	Quantum and Statistical Physics, 1st-year engineering (BSc 3), CentraleSupélec
Dec. 2024	10 h TD equiv.	General Physics, BSc Mathematics & Computer Science, Université Paris Cité
Oct.–Dec. 2024	30.5 h TD equiv.	Optics, BSc Biomedical Sciences, Université Paris Cité
Mar.–Jun. 2024	13.5 h TD equiv.	Quantum and Statistical Physics, 1st-year engineering (BSc 3), CentraleSupélec
2015–2024	25–250 h/year	Private tutoring in mathematics and physics, secondary school to first-year bachelor's

Science Outreach

- Dec. 2025 Outreach text produced for the *LIED Manifesto* (lab's popular science book).
- 5 Dec. 2025 Science discovery session designed for Year 2 primary school pupils.
- 20 Dec. 2024 *Art & Mathématiques* conference aimed at artists: visual introduction to network theory and the energy transition.
- 8 Oct. 2023 Introduction to chaotic processes, *Fête de la Science*, ENS Paris-Saclay.
- 2021–2022 Member of *Cours aux Ernest*: introductory physics lectures for ENS Paris students.
- 2021 Contribution to a science outreach article in the *Huffington Post*: *Le muon, particule révolutionnaire qui agite le monde de la physique*.
- 2019 Articles in the *Alma Mater* journal: *La théorie des mondes parallèles : de l'autre côté de l'horizon cosmologique* and *Un aperçu des abysses de l'univers*.

Other Training

Philosophy — registered auditor in the **LOPHISS-SPH** Master's programme (Philosophy and History of Science), Université Paris Diderot, 2018–2019

Pedagogy — training in the cognitive sciences of learning (MOOCs, doctoral school courses).

Other — alto saxophone (conservatory and Arpej Jazz School, 2018–2019); PSC1 first-aid certificate; EU driving licence (category B).